MTH1002

UNIVERSITY OF EXETER

COLLEGE OF ENGINEERING, MATHEMATICS AND PHYSICAL SCIENCES

MATHEMATICS

May 2020

Mathematical Methods

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You have 24 hours to complete this paper from the time of its release
Intended duration: 2 hours

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is an **OPEN BOOK** examination

SECTION A

1. (a) Find the following indefinite integrals

$$\text{(i)} \int \left(\cos\frac{x}{2} - \sin\frac{x}{2}\right)^2 dx, \quad \text{(ii)} \int \frac{dx}{\sqrt{x^2-4}}. \tag{10}$$

(b) Express the definite integral

$$\int_0^3 (4x+1)\,dx$$

as a Riemann sum and thus evaluate it. Verify the answer by calculating the integral using rules of integration. (5)

(c) Evaluate the following definite integrals by using suitable substitutions or otherwise

$$\text{(i)} \int_0^5 \frac{t\,dt}{\sqrt{1+t^2}}, \quad \text{(ii)} \int_0^5 x^2\sqrt{25-x^2}\,dx. \tag{10}$$

(d) Solve the following ordinary differential equations

(i) $\dfrac{dy}{dx} + \dfrac{3}{x}y = 4 + 6x^2$ with $y(1) = 2$,

(ii) $\dfrac{d^2y}{dx^2} + y = e^{-x}$ with $y(0) = 1$ and $y'(0) = -1$. (10)

(e) Suppose that $f(x) = x^2 + 3x + 6$ and that the matrix $A = (a_{ij})_{2\times 2}$ is given by

$$A = \begin{bmatrix} -1 & 0 \\ 4 & 3 \end{bmatrix}.$$

Evaluate $f(A)$. (5)

(f) Evaluate the following integrals

$$\text{(i)} \int_0^1 \int_0^1 3(x^2+y^2)\,dy\,dx, \quad \text{(ii)} \int_0^{\pi/4} \int_{\sin\theta}^1 r\sin\theta\,dr\,d\theta. \tag{10}$$

[**50**]

SECTION B

2. (a) Find all the eigenvalues of the matrix $A = (a_{ij})_{5\times 5}$ given by

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

Hint: Make use of the elementary row operation. (15)

(b) Evaluate the following sum by associating it with a Riemann sum and hence a definite integral

$$I = \lim_{n\to\infty} \left[\frac{1^{11} + 2^{11} + 3^{11} + ... + n^{11}}{n^{12}} \right].$$

(10)

[**25**]

3. (a) Find the values of a and b in

$$\xi = x + ay, \quad \eta = x + by$$

such that they transform the partial differential equation

$$\frac{\partial^2 u}{\partial x^2} + 5\frac{\partial^2 u}{\partial x \partial y} + 6\frac{\partial^2 u}{\partial y^2} = 0$$

into

$$\frac{\partial^2 u}{\partial \xi \partial \eta} = 0.$$

Write down the general solution of the equation. (15)

(b) Suppose that the square matrix $A = (a_{ij})_{3\times 3}$ is symmetric ($a_{ij} = a_{ji}$) with $a_{11} = a_{22} = a_{33} = 0$ and has an eigenvalue 2 with the corresponding eigenvector $\mathbf{v} = [1, 2, -1]^T$. Find the matrix A. (10)

[**25**]

4. (a) Find the general solution of the ordinary differential equation

$$\frac{d^2x}{dt^2} + x = 1 - \frac{1}{\sin t}.$$

Hint: Consider the particular integral in the form

$$x_{PI}(t) = C_1(t)\cos t + C_2(t)\sin t$$

and, then, derive $C_1(t)$ and $C_2(t)$ by assuming that

$$\frac{dC_1}{dt}\cos t + \frac{dC_2}{dt}\sin t = 0.$$

(15)

(b) Suppose that $f(x)$ is continuous and satisfies the equation

$$\int_0^x \xi f(x-\xi)d\xi = \sin^2(2x).$$

Find $f(x)$. (10)

[**25**]